

**SET-5****Series %BAB%**प्रश्न-पत्र कोड
Q.P. Code**56/B/5**

रोल नं.

Roll No.

--	--	--	--	--	--	--	--

परीक्षार्थी प्रश्न-पत्र कोड को उत्तर-पुस्तिका के मुख-पृष्ठ पर अवश्य लिखें।

Candidates must write the Q.P. Code on the title page of the answer-book.

- कृपया जाँच कर लें कि इस प्रश्न-पत्र में मुद्रित पृष्ठ 11 हैं।
- प्रश्न-पत्र में दाहिने हाथ की ओर दिए गए प्रश्न-पत्र कोड को परीक्षार्थी उत्तर-पुस्तिका के मुख-पृष्ठ पर लिखें।
- कृपया जाँच कर लें कि इस प्रश्न-पत्र में 12 प्रश्न हैं।
- कृपया प्रश्न का उत्तर लिखना शुरू करने से पहले, उत्तर-पुस्तिका में प्रश्न का क्रमांक अवश्य लिखें।
- इस प्रश्न-पत्र को पढ़ने के लिए 15 मिनट का समय दिया गया है। प्रश्न-पत्र का वितरण पूर्वाह्न में 10.15 बजे किया जाएगा। 10.15 बजे से 10.30 बजे तक छात्र केवल प्रश्न-पत्र को पढ़ेंगे और इस अवधि के दौरान वे उत्तर-पुस्तिका पर कोई उत्तर नहीं लिखेंगे।
- Please check that this question paper contains 11 printed pages.
- Q.P. Code given on the right hand side of the question paper should be written on the title page of the answer-book by the candidate.
- Please check that this question paper contains 12 questions.
- Please write down the serial number of the question in the answer-book before attempting it.
- 15 minute time has been allotted to read this question paper. The question paper will be distributed at 10.15 a.m. From 10.15 a.m. to 10.30 a.m., the students will read the question paper only and will not write any answer on the answer-book during this period.



रसायन विज्ञान
(केवल दृष्टिबाधित परीक्षार्थियों के लिए)

**CHEMISTRY****(FOR VISUALLY IMPAIRED CANDIDATES ONLY)**

निर्धारित समय : 2 घण्टे

Time allowed : 2 hours

अधिकतम अंक : 35

Maximum Marks : 35



General Instructions :

Read the following instructions very carefully and strictly follow them :

- (i) *This question paper contains **12** questions. **All** questions are compulsory.*
- (ii) *This question paper is divided into **three** Sections – Section **A**, **B** and **C**.*
- (iii) ***Section A** – Questions no. **1** to **3** are very short answer type questions, carrying **2** marks each.*
- (iv) ***Section B** – Questions no. **4** to **11** are short answer type questions, carrying **3** marks each.*
- (v) ***Section C** – Question no. **12** is a case based question, carrying **5** marks.*
- (vi) *Use of log tables and calculators is **not** allowed.*

SECTION A

- 1. Write the chemical equations involved when aniline is treated with the following reagents : (Any **two**) $2 \times 1 = 2$
 - (i) CH_3COCl /Pyridine
 - (ii) Br_2 water
 - (iii) HCl
- 2. Define electrochemical cell. What happens if external potential applied becomes greater than E_{cell}^0 ? **2**
- 3. Write two differences between Order of reaction and Molecularity of reaction. $2 \times 1 = 2$

SECTION B

- 4. Account for the following : $3 \times 1 = 3$
 - (i) Transition metals generally form coloured compounds.
 - (ii) Zn has the lowest enthalpy of atomisation.
 - (iii) Atomic radii of 5d and 4d series of elements are almost identical.



5. (a) (i) Define the following terms with an example :
(I) Ambidentate ligand
(II) Chelate effect
(ii) Write the formula of the following complex :
Hexaammineplatinum(II)chloride 2+1=3

OR

- (b) (i) Write the hybridization and geometry of $[\text{Ni}(\text{CN})_4]^{2-}$.
(Atomic number of Ni = 28)
(ii) Write the IUPAC name of the following complex :
 $[\text{Co}(\text{NH}_3)_5\text{Cl}]\text{Cl}_2$
(iii) Why are low spin configurations rarely observed for tetrahedral coordination entities ? 3×1=3
6. (a) Define the following terms with an example of each : 3×1=3
(i) Lyophilic sol
(ii) Gel
(iii) Multimolecular colloid

OR

- (b) Write three differences between Physisorption and Chemisorption. 3×1=3
7. (a) Write the reactions involved in the following : 3×1=3
(i) Hoffmann bromamide degradation reaction
(ii) Gabriel phthalimide synthesis
(iii) Carbylamine reaction

OR

- (b) Write the structures of A, B and C in the following : $2 \times 1 \frac{1}{2} = 3$
- (i) $\text{CH}_3\text{COOH} \xrightarrow[\Delta]{\text{NH}_3} \text{A} \xrightarrow[\text{H}_2\text{O}]{\text{LiAlH}_4} \text{B} \xrightarrow[0^\circ\text{C}]{\text{HNO}_2} \text{C}$
- (ii) $\text{C}_6\text{H}_5\text{NO}_2 \xrightarrow{\text{Fe} + \text{HCl}} \text{A} \xrightarrow[453 - 473 \text{ K}]{\text{H}_2\text{SO}_4} \text{B} \xrightarrow{\quad} \text{C}$



8. Calculate the emf of the following cell : 3



Given : $E_{\text{cell}}^0 = 1.56 \text{ V}$,

$$[\log 2 = 0.3010, \log 4 = 0.6021, \log 10 = 1]$$

9. A first order reaction is 25% complete in 30 minutes. Calculate the value of rate constant and $t_{1/2}$. 3

$$[\log 2 = 0.3010, \log 4 = 0.6021, \log 3 = 0.4771]$$

10. (a) Give reasons for the following : $3 \times 1 = 3$

- (i) In semicarbazide, there are two $-\text{NH}_2$ groups, but only one is involved in the formation of semicarbazones.
- (ii) Carboxylic acid is a stronger acid than phenol.
- (iii) α -hydrogens of aldehydes and ketones are acidic in nature.

OR

- (b) An organic compound (A) with molecular formula $\text{C}_5\text{H}_{10}\text{O}$ does not reduce Tollens' reagent, but forms an addition compound with NaHSO_3 and gives positive iodoform test on heating with NaOH in the presence of I_2 . On vigorous oxidation, it gives ethanoic and propanoic acid.

Write

- (i) the possible structure of the compound.
- (ii) the IUPAC name of the compound.
- (iii) the reaction of (A) with NaHSO_3 . $3 \times 1 = 3$



11. (i) Which metal in the first series of transition metals exhibits +1 oxidation state most frequently and why ?
- (ii) Of the d^4 species, Cr^{2+} is strongly reducing while Mn^{3+} is strongly oxidising, why ?
- (iii) Name a transition element in 3d series
- (I) which does not exhibit variable oxidation states, and
- (II) which shows a larger number of oxidation states.

$3 \times 1 = 3$

SECTION C

12. Read the passage given below and answer the questions that follow :

Aldehydes differ from ketones in their oxidation reactions. Aldehydes are easily oxidised to carboxylic acids on treatment with common oxidising agents like nitric acid, $KMnO_4$, $K_2Cr_2O_7$, etc. Even mild oxidising agents mainly Tollens' reagent and Fehling solution also oxidise aldehydes. Ketones are generally oxidised under vigorous conditions.

Aldehydes and ketones undergo nucleophilic addition reactions onto the carbonyl group with a number of nucleophiles such as HCN , $NaHSO_3$, alcohols, ammonia derivatives and Grignard reagents. The α -hydrogens of aldehydes and ketones are acidic due to which they undergo Aldol condensation.

- (i) Why is the oxidation of Aldehydes easier than Ketones ?

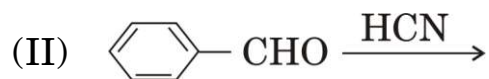


- (ii) Arrange the following compounds in the increasing order of their reactivity towards nucleophilic addition reaction :

Ethanal, Acetone, Propanal, Acetophenone

- (iii) Write a simple chemical test to distinguish between Acetone and Ethanal.

- (iv) (a) Write the major product in the following reactions : 1+1+1+2=5



OR

- (b) How will you bring about the following conversions ?

(I) Benzoic acid to Benzaldehyde

(II) Propan-2-ol to Propanone